Variant Interpretation on CYP2C9

BIOT 6900 · Module 5 Assignment

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Background

Individuals may react in different ways to the same medicine and dosage, and one reason for that is their slight differences in genetic makeup of enzymes which work on metabolizing that medicine in their body. The field of pharmacogenomics uses the connection between genetics and drug response to make predictions about a certain person's reaction to a certain medicine based on their own genetic data. There are several computational approaches that could be used for prediction of whether the variation is functionally significant or not, and CADD, REVEL, and Alpha Missense are some of them, which use their own approach for scoring functional impact of a genetic variant, while the AlphaGenome Atlas tries to detect those effects which have been missed by other methods, especially in regions of the genome that do not code for a protein at all. In this paper, these computational methods are being applied to a single gene, CYP2C9, for assessing their capabilities and limitations.

Overview

CYP2C9 is the name of the liver enzyme responsible for breaking down warfarin, which is a blood thinner medication. An enzyme that works slower than normal causes more warfarin to stay in the body longer than intended and thus leads to a standard dose becoming an overdose, putting someone at higher risk for actual bleeding. The choice of CYP2C9 was made because the two well-known low-activity alleles of this gene are both simple amino acid substitutions, which allows most computational tools to detect and assess them while other pharmacogenes have their functional alleles located beyond the reach of protein-centered tools. I extracted 41 variants from the open database; both reduced activity alleles appeared immediately, suggesting that my search worked properly and scoring coverage was very high: CADD covered 93 percent, REVEL and AlphaMissense covered 81 percent. AI assistance was used to write the needed Python code for extracting and processing the data; the rest is my own opinion and threshold values I chose.

Summary of key results:

Metric	Value
Variants fetched	41 (both anchor alleles *2, *3 found)
Score coverage	CADD 92.7%, REVEL 80.5%, AlphaMissense 80.5%
Frequency filter	MAX_AF = 0.10 (0.001 would discard 4 drug-response variants)
Calibration	CADD, 8 labelled variants, flagged unreliable
VUS classified	22 total, exported to pgx_classified_CYP2C9.csv
AlphaGenome Atlas	36 variants scored live

This is the pipeline, start to finish:



1. Frequency cutoff and calibration

CYP2C9's two defining alleles sit at roughly 5 percent and 9 percent of the population, common because having a slightly slower drug-clearing enzyme was never something evolution had any reason to punish, not until humans began manufacturing warfarin. I set my cutoff at 10 percent for exactly this reason, since a standard, much stricter cutoff would have erased four real clinical variants outright, including both of this gene's famous altered alleles, the difference between a pipeline that actually works for this gene and one that silently throws away its two most important answers.

I calibrated CADD rather than REVEL since it covered more of my variants, but the result cannot be trusted as real evidence. It rests on only 8 labelled variants total, and the likelihood ratios even ran backwards, lower scores looked more concerning than higher ones, a clear sign of too little data rather than real biology speaking. When I reclassified my 22 uncertain variants against this shaky calibration, 18 of them technically moved toward one side or the other, but none of those shifts should actually be acted on, since they only inherit the same underlying weakness. CYP2C9 simply lacks enough confirmed, well-studied variants to calibrate any predictor meaningfully, despite having excellent data coverage otherwise.

CADD band	n	P/LP	B/LB	LR+
0–10	4	1	3	2.33
10–20	1	0	1	0.00
20–25	3	0	3	0.00
25–30	0	–	–	undefined
30–50	0	–	–	undefined

Every band rests on fewer than ten variants (at most four), and the whole calibration on just 1 P/LP and 7 B/LB. I chose not to pool with a second gene, so I am disclosing that calibration is not possible for CYP2C9 with current ClinVar labels. The 22 VUS split 9 leans pathogenic, 9 leans benign and 4 stay VUS, but none of those moves should be acted on.

2. Coverage gaps and the regulatory genome

CADD, REVEL and AlphaMissense could not score the single frameshift variant in my gene at all (ClinVar 285601). Interestingly, ClinVar lists it as benign or likely benign. It sits in a splice region, and a frameshift near the very end of a gene can escape the cell's mRNA-decay machinery and leave a mostly working protein, so a blank score here definitely does not mean "benign" or "pathogenic". An RNA assay (RT-PCR on cells carrying the variant) would show whether the transcript is decayed or translated, and an enzyme activity assay would show whether the protein still works.

The Atlas's top-scoring variant, rs370407644 (AVI 23.4), is labelled missense, but its score was driven by MERGED_SPLICING, which suggests it may act by disrupting splicing rather than by changing the amino acid. REVEL scored it low (0.14), which fits, since REVEL only sees the protein change. An RNA splicing assay would settle it. For most other variants, ALPHAMISSENSE drove the score, meaning the Atlas mostly re-found coding effects my missense tools had already seen.

3. A disagreement worth reading, not explaining away

The variant I picked is rs28371685, which changes arginine 335 to tryptophan (R335W) and is better known as CYP2C9*11. It tripped the CADD-vs-REVEL rule: CADD gave it a high 27.5, but REVEL (0.22) and AlphaMissense (0.24) both scored it low. At first that looks like the tools just disagree, but I think both are telling the truth. REVEL and AlphaMissense were built to catch variants that cause disease, and this one doesn't. Nobody gets sick from carrying it

. It only matters when that person takes warfarin, because the enzyme works a bit slower and the drug builds up. So to those two tools, it looks harmless. CADD looks at something different: how much this position in the gene has stayed the same across species, and position 335 has barely changed, so CADD flags it. My reading is that this is a real but mild loss of function that only shows up

when a drug is involved. The ClinVar “drug response” label fits that, and so does its 0.7% frequency. If it were truly damaging, evolution would have kept it much rarer.

4. Limitations

I have no direct evidence of which direction this variant actually pushes enzyme activity, faster or slower, only that it looks statistically unusual on two scales that were never built to measure enzyme speed directly. I also cannot say whether it should change a real patient's actual warfarin dose, a predictor disagreement is a reason to investigate further, not a diagnosis to act on.

Moreover, it is impossible to know whether this variant would affect the dosage of warfarin a real-world patient might take; a disagreement about the prediction is precisely what needs investigating further and not something upon which to base a course of treatment. The experiment that would actually settle this is simple to describe: make the R335W version of the enzyme in the lab, put it next to the normal enzyme, and measure how fast each one breaks down warfarin. If R335W turned out to work just as fast as normal, I would change my mind. That would mean CADD was only reacting to the position being conserved, and this variant doesn’t really matter for warfarin dosing.

REF:

1. Johnson, J. A., et al. (2017). Clinical Pharmacogenetics Implementation Consortium (CPIC) guideline for pharmacogenetics-guided warfarin dosing: 2017 update. Clinical Pharmacology & Therapeutics, 102(3), 397–404.